

necessary to give our mission the momentum, but the ultimate goal must be reached by homegrown chip companies.

It is a widely known fact that making chips is a precarious business, which needs steep investment. It becomes even more tedious when a new startup wants to enter the space thanks to the high cost of entry. “The first challenge these startups face is the cost of EDA tools which are prohibitively expensive. Almost 25% of the total cost of the entire design chain is attributed to the cost of the EDA tools,” says Vivek Khaneja, Executive Director, C-DAC.

And although this is being addressed to some extent by open source tools and open domain companies, which help provide easier access to EDA tools and IPs, the truth is that their standard offering is beyond the reach of a small startup.

Availability of IPs is another major obstacle that prevents chip startups from moving forward. When you are trying to solve a problem, you will create a solution that has some sort of programmability and peripheral IPs. Getting hold of those IPs, which are silicon proven today, is not easy and, most importantly, it is not cheap.

Open source IPs have made life easier but only on the surface. At the crux of it, the quality of such IPs may be subpar and integrating those IPs in the design is a matter of shooting in the dark with the hope of striking Eureka.

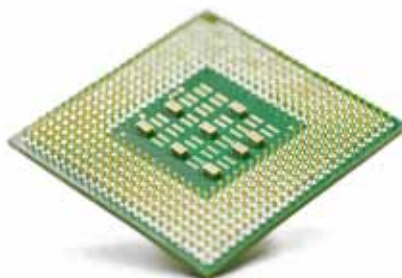
Additionally, building a prototype for a chip startup is extremely difficult due to their limited funding and the fact that, typically, foundries refrain from giving them much leverage when it comes to prototyping.

Mike Wishart, CEO, e-Fabless, points out, “You look at what happened in software, by removing all the upfront cost, they took out the ‘no’ out of innovation. One of the

biggest issues you get in silicon is that there are so many things you have to spend money on.”

Funding for chip startups

“Based on my experience of raising money for my own company, a big part of getting companies funded and up and running is, in fact, persistence,” explains Wishart. Hardware startups are notorious for being incapable of catching the eye of investors due to the high investments that they require along with the even higher risks and marginally lower profits associated with them.



When you come back to the life cycle of funding a company, it is a ‘crawl-walk-run’ cycle for every company. The crawl part is when you have to get the proof of concept, and then come the walk and run part when you produce it and then scale it up. “The funding difference between software and semiconductor is not that different in the walk and run part but it’s the crawl, the getting started part, which is the killer,” says Wishart.

The existing semiconductor ecosystem is built and consolidated around creating and making standard product chips for mass markets, where the basic model is to pour tremendous amounts of money into capital equipment and R&D, create a product and sell a lot of them to enable PCs and phones. Industry leaders emerged in areas like EDA, IP, and foundry and they quite understandably protected their investment licenses that must be signed and often paid for upfront. As a result, not only did it get more expensive to

start a company, but it also became incredibly time consuming.

The emergence of open source led to a great amount of cost-reduction along with simplification, sharing and faster time to design. Yet, their delivery for chips that are optimised for quality, power, performance, and cost is often not at the level required for startups to be able to scale their product. It gives the required push, but it may not be sufficient to keep the momentum.

Building a company from the ground up is a tedious task. As such, building a chip startup is difficult on its own. Moreover, the chip ecosystem is an extremely rigid one that leaves very little space for newcomers. How do we turn the tide?

“I think the tide will only be turned if there is a demand for locally produced chips,” says Sherwani. While the government’s efforts to support domestic startups to encourage them to build chips is commendable, there is a need for these chips to come to the forefront, which can only happen when enough demand is created.

Sherwani suggests having provisions that may help incentivise the use of locally-made chips.

“I’m not saying putting a barrier for big players, but indigenous players need to be given more benefits. As part of this overwhelming plan, we must create incentives for startups,” Sherwani adds.

Lack of VC infrastructure

The lack of a proper venture capitalist (VC) or venture capital infrastructure is another obstacle in allowing proper funding to flow in the right direction when it comes to hardware startups. A lot of VCs that exist in the country do so only on the surface level. At the core, these are just US based VCs who operate in India, among other places. The need, in this case, is much more precise—India needs Indian VCs—to understand the local demand